

Random Numbers

For Projects

What Are Random Numbers?

- Random numbers are arbitrary numbers
- No one can predict the number as its random!
- Used in games and lotteries
- More details on Python Random numbers (3.x branch)
  - https://docs.python.org/3/library/random.html

Simple Program to Test Random Values

```
import random

# Create variables
rand = 0

# Generate numbers till 10
rand = random.randrange(10)
print(rand)

# Generate numbers from 1 to 50
rand = random.randrange(1,50)
print(rand)

# Shuffel values in a list
numbers = [1,2,3,4,5,6,7,8,9,10]
print(numbers)
random.shuffle(numbers)
print(numbers)

# Generate next random floating point 0.0 ~ 1.0
rand = random.random()
print(rand)
```

Program Output

```
3
14
[1, 2, 3, 4, 5, 6, 7, 8, 9, 10]
[3, 6, 2, 7, 4, 8, 1, 5, 9, 10]
0.6924968261686208
```

```
9
44
[1, 2, 3, 4, 5, 6, 7, 8, 9, 10]
[2, 9, 5, 8, 4, 7, 1, 3, 6, 10]
0.9109411910662312
```